



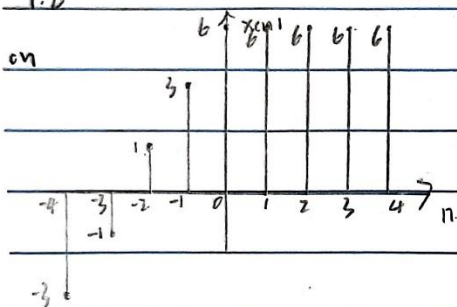
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2024.09.22

1.1

$$x(n) = \delta(n+4) + 2\delta(n+2) - \delta(n+1) + 2\delta(n) + \delta(n-1) \\ + 2\delta(n-2) + 4\delta(n-3) + \frac{1}{4}\delta(n-4) + 2\delta(n-6)$$

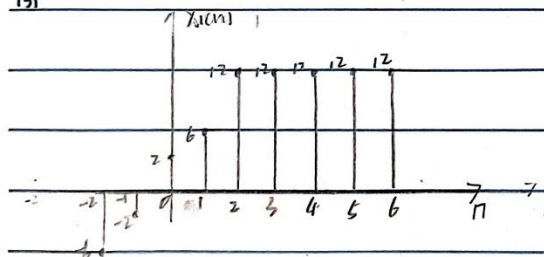
1.2



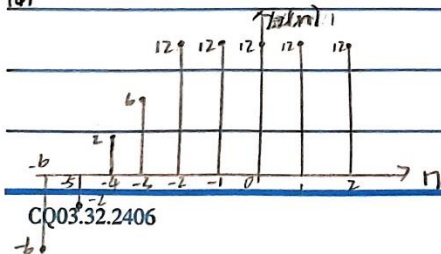
(2)

$$x(n) = -\frac{1}{2}\delta(n+4) - \delta(n+3) + \delta(n+2) + \frac{1}{2}\delta(n+1) \\ + b\delta(n) + b\delta(n-1) + b\delta(n-2) + b\delta(n-3) + b\delta(n-4) \\ = \sum_{m=-4}^{-1} (2m+5)\delta(n-m) + \sum_{m=0}^4 b\delta(n-m)$$

(3)



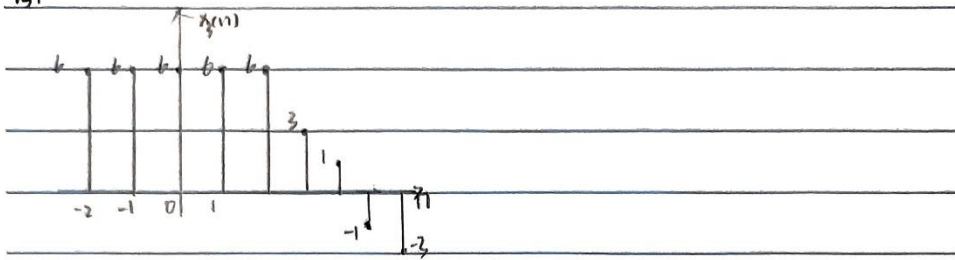
(4)



CQ03.32.2406

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1.51



1.52

$$m \quad w = \frac{3}{7}w$$

$$T = \frac{2\pi}{w}k = \frac{14}{3}k \quad \text{for } k=3 \text{ 时 } T=14.$$

$\therefore x[n]$ 是一个 $T=14$ 的周期序列.

$$(2) \quad x[n] = \cos\left(\frac{1}{8}n - \frac{\pi}{4}\right) + j \sin\left(\frac{1}{8}n - \frac{\pi}{4}\right)$$

$$w = \frac{1}{8} \quad T = \frac{2\pi}{w}k = 16\pi k$$

$\therefore T$ 为无理数

$\therefore x[n]$ 不是一个周期序列.

1.5.

$$(2) \quad y(n) = 2x(n) + 3.$$

线性判断: 先组合再系统 先系统后组合.

$$2(ax_1(n) + bx_2(n)) + 3 \neq ay_1(n) + by_2(n)$$

时变判断 先用移的系统 先系统后时移.

$$y(n-n_0) = 2x(n-n_0) + 3$$

则该系统是非线性时不变系统. 非线性时不变.

$$(3) \quad y(n] = x[n-n_0]$$

$$\therefore ay_1(n) + by_2(n) = ax_1[n-n_0] + bx_2[n-n_0]$$

$$x[n-n_0-n_1] = y[n-n_1]$$

则该系统为线性时不变系统.

$$(4) \quad y(n) = x(n)$$

$$ay_1(n) + by_2(n) = ax_1(n) + bx_2(n)$$

$$x(-n+n_0) \neq y(n-n_0) = x(-n-n_0)$$

则该系统为线性时变系统.



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1.b.

$$m \quad y(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(n-k)$$

因果性: $y(0) = \frac{1}{N} \sum_{k=0}^{N-1} x(k)$ 仅与该时刻, 以及以前的值关联, 只要 $N \neq 1$.

稳定性: $|x(n)| \leq M, |n| |y(n)| \leq M.$

$$12) \quad y(n) = x(n) + x(n+1)$$

因果性: n 时刻的值与 $n+1$ 时刻的输入有关, 则非因果

稳定性 若 $|x(n)| \leq M, |n| |y(n)| \leq 2M.$

$$13) \quad y(n) = \sum_{k=n-n_0}^{n+n_0} x(k)$$

因果性: 若假设 $n_0 > 0$, 系统为非因果. 当且仅当 $n_0 = 0$ 该系统因果.

稳定性: 若 $|x(n)| \leq M, |n| |y(n)| \leq (|2n_0| + 1)M.$

$$14) \quad y(n) = x(n-n_0)$$

因果性: 若 $n_0 \leq 0$, 系统非因果 若 $n_0 > 0$, 系统因果

稳定性: 若 $|x(n)| \leq M, |n| |y(n)| \leq M$

$$15) \quad y(n) = e^{x^2(n)}$$

因果性: n 时刻值, 仅与 $x(n)$ 在 n 时刻值关联, 则为因果系统

稳定性: $|x(n)| \leq M, |n| |y(n)| \leq e^{M^2}$



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1.7.

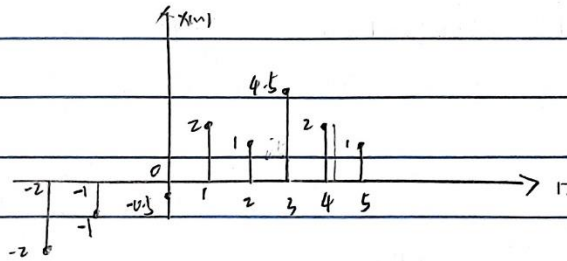
$$h(n) = 2\delta(n) + \delta(n-1) + 0.5\delta(n-2)$$

$$x(n) = -\delta(n+2) + \delta(n-1) + 2\delta(n-3)$$

$$|a| \quad y(n) = x(n) * h(n)$$

$$= -2\delta(n+2) + 2\delta(n-1) + 4\delta(n-3) - \delta(n+1) + \delta(n-2) + 2\delta(n-4)$$

$$+ 0.5\delta(n) + 0.5\delta(n-3) + \delta(n-5)$$

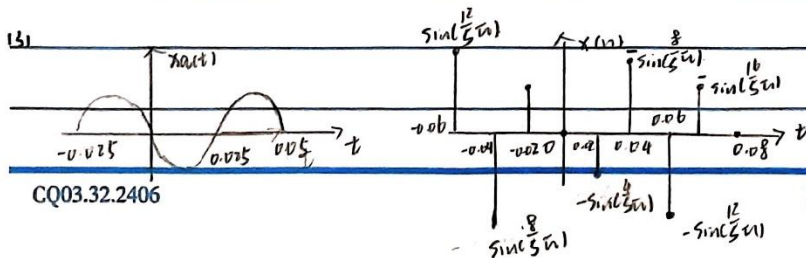


1.13.

$$c_n. \quad x_a(t) = \cos(2\pi f t + \frac{\pi}{4}) = -\sin(2\pi f t)$$

$$T = \frac{1}{f} = \frac{1}{20} = 0.05 \text{ s}$$

$$(2) \quad \hat{x}_a(t) = \sum_{n=-\infty}^{\infty} x_a(nT) \delta(t-nT) \quad T=0.02 \text{ s}$$



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$$x(n) = \sum_{n=-\infty}^{\infty} \sin(2\pi f n T) \quad T = 0.02s$$

$$T' = \frac{2\pi}{2\pi f T} K = \frac{5}{K} \quad K=2 \text{ 时}, T'=5$$

则其周期为5